

CSE332

INDUSTRY ETHICS AND LEGAL ISSUES

**Hands on tools for
cyber security), case studies on
cyber crimes)**

The Growing Importance of Cybersecurity

- ✓ Cybersecurity is more critical than ever.
- ✓ By 2025, global damage from cybercrime is expected to reach **\$10.5 trillion**.
- ✓ **Ransomware attacks** rose by **13% in 2023**.
- ✓ An estimated **68% of business leaders** feel their cybersecurity risks are increasing.

❑ Learn how to stay secure.



Financial impact

Economic losses



Ransomware increase

Growing threat



Increasing risks

Business concerns



Essential Cybersecurity Tools for Protection

- Use these essential tools to enhance your cybersecurity:
 - ✓ **Firewalls** provide a first line of defense.
 - ✓ **VPNs** protect data in transit by encrypting connections.
 - ✓ **Anti-virus software** prevents known threats.
 - ✓ **Intrusion Detection Systems (IDS)** provide real-time threat monitoring.

Cybersecurity Tools

1. Network Security Tools:
 - **Wireshark, Nmap, Snort, Tcpdump**
2. Endpoint Security Tools:
 - **CrowdStrike Falcon, Symantec, Bitdefender GravityZone**
3. Penetration Testing Tools:
 - **Metasploit, Burp Suite, Nessus, Kali Linux**
4. Forensics & Investigation Tools:
 - **Autopsy, FTK Imager, Volatility**
5. SIEM Tools:
 - **Splunk, IBM QRadar, ELK Stack**

Advanced Tools for Proactive Defense

- Employ advanced tools for robust defense:
 - ✓ **Endpoint Detection and Response (EDR)** identifies and responds to endpoint threats.
 - ✓ **Vulnerability scanners** detect system weak points.
 - ✓ **Encryption tools** secure sensitive data.
 - ✓ **Security Information and Event Management (SIEM)** offers holistic monitoring.

Case Study 1: Colonial Pipeline Ransomware Attack (2021)

- ✓ The **Colonial Pipeline attack** disrupted the U.S. fuel supply.
- ✓ Hackers received a **\$4.4 million Bitcoin payment**.

This case highlights the importance of:

- ✓ **Multi-Factor Authentication (MFA)**
- ✓ **Network segmentation**
- ☐ Take steps now to ensure you are secure.

Case Study 2: Email Phishing Against a Fortune 500 Company

- ✓ A **phishing attack** led to a **\$100 million loss** in wire fraud.
- ✓ The attack vector involved **deceptive emails impersonating executives**.

To prevent email scams, implement:

- ✓ **Staff training**
- ✓ **SPF, DKIM, and DMARC** email authentication protocols



Case Study 3: Data Breach at Equifax (2017)

- ✓ The **Equifax data breach** compromised personal data of **147 million people**.
- ✓ Cause: An **unpatched vulnerability** in **Apache Struts** software.

This case illustrates the critical need for:

- ✓ **Timely software updates**
- ✓ **Effective patch management**



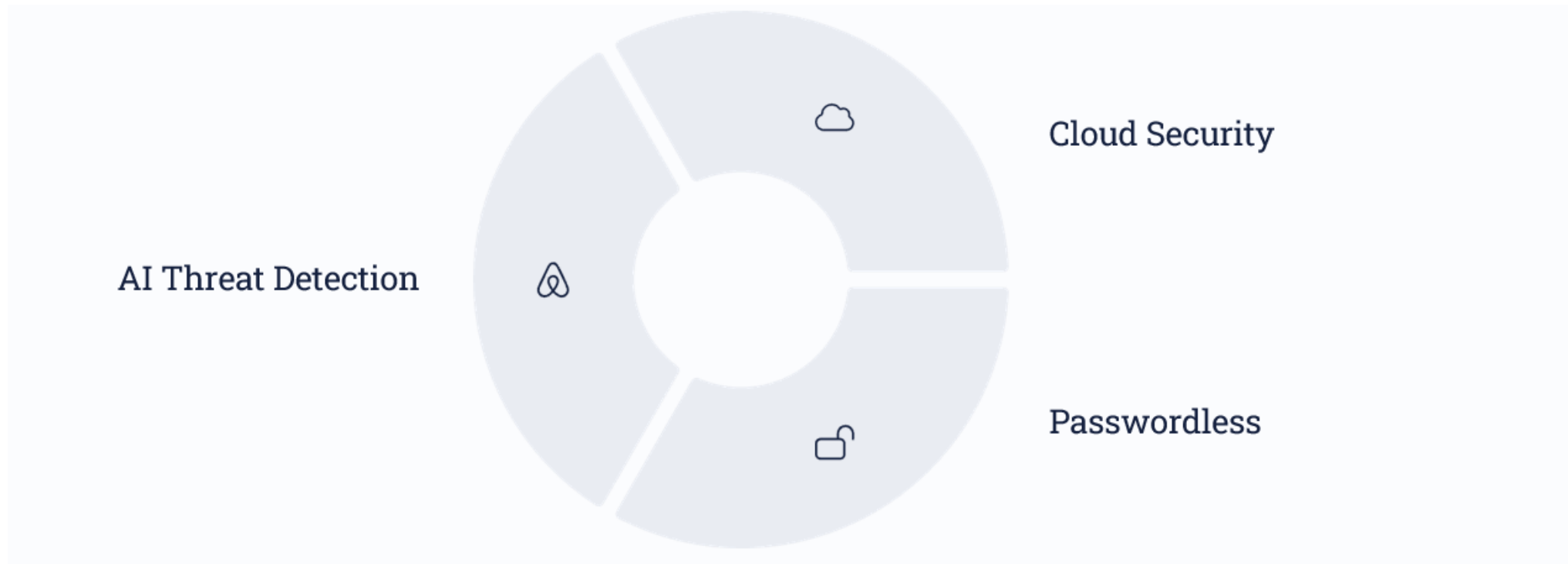
Best Practices for Cybersecurity Resilience

- ✓ Regular software updates and patch management
- ✓ Implement Multi-Factor Authentication (MFA)
- ✓ Conduct employee cybersecurity awareness training
- ✓ Use strong, unique passwords and a password manager
- ✓ Backup critical data regularly and securely
- ✓ Utilize firewalls, antivirus, and endpoint protection
- ✓ Perform regular vulnerability assessments and penetration testing
- ✓ Enforce least privilege access controls
- ✓ Deploy network segmentation and monitoring tools
- ✓ Establish a robust incident response and recovery plan



Emerging Trends in Cybersecurity Tools

1. Stay ahead with emerging tools. AI-powered threat detection offers predictive analytics.
2. Secure Access Service Edge (SASE) enhances network security.
3. Password less authentication uses biometrics. Cloud-focused tools protect multi-cloud environments.



Conclusion: Staying Ahead in Cybersecurity

- **Cybersecurity is continuous, not a one-time fix. Combine tools, processes, and trained people.**
 - **Adopt new tools, upgrade existing systems, and educate your team to stay ahead.**
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